KAIZEN MATHS

Lesson starter

Advanced Matrices: Transformations and Eigenvalues

Matrix Transformations

Name _____

S1

Find the image of each point under the matrix transformation.

$A = \begin{bmatrix} 0 & -2 \\ 1 & -3 \end{bmatrix}$, $P = \begin{bmatrix} -2 \\ -2 \end{bmatrix}$

S2

Find the combined matrix for each pair of transformations.

A = $\begin{bmatrix} 2 & -2 \\ 1 & 3 \end{bmatrix}$, B = $\begin{bmatrix} 2 & 1 \\ 2 & 3 \end{bmatrix}$, apply A then B

S3

Find the fixed line for each transformation.

$A = \begin{bmatrix} 1 & -2 \\ 0 & 4 \end{bmatrix}$

S4

Find the image of each point under the matrix transformation.

$A = \begin{bmatrix} 0 & -2 \\ 0 & 1 \end{bmatrix}$, $P = \begin{bmatrix} -4 \\ -2 \end{bmatrix}$

Show your working

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S1

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$A = \begin{bmatrix} 0 & -2 \\ 0 & 1 \end{bmatrix}$, $P = \begin{bmatrix} -4 \\ -2 \end{bmatrix}$

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KAIZEN MATHS

Exit ticket

Advanced Matrices: Transformations and Eigenvalues

Matrix Transformations

Name _____

EA1

Find the combined matrix for each pair of transformations.

A = $\begin{bmatrix} 2 & 4 \\ 3 & 4 \end{bmatrix}$, B = $\begin{bmatrix} 3 & 1 \\ 1 & 0 \end{bmatrix}$, apply A then B

EA2

Find the fixed line for each transformation.

$A = \begin{bmatrix} 1 & 4 \\ 0 & 4 \end{bmatrix}$

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Exit ticket

Advanced Matrices: Transformations and Eigenvalues

Matrix Transformations

Name _____

EB1

Find the image of each point under the matrix transformation.

$A = \begin{bmatrix} 1 & 0 \\ -2 & -3 \end{bmatrix}$, $P = \begin{bmatrix} -2 \\ 5 \end{bmatrix}$

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Exit ticket

Advanced Matrices: Transformations and Eigenvalues

Matrix Transformations

Name _____

EA1

Find the combined matrix for each pair of transformations.

A = $\begin{bmatrix} 2 & 4 \\ 3 & 4 \end{bmatrix}$, B = $\begin{bmatrix} 3 & 1 \\ 1 & 0 \end{bmatrix}$, apply A then B

EA2

Find the fixed line for each transformation.

$A = \begin{bmatrix} 1 & 4 \\ 0 & 4 \end{bmatrix}$

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Exit ticket

Advanced Matrices: Transformations and Eigenvalues

Matrix Transformations

Name _____

EB1

Find the image of each point under the matrix transformation.

$A = \begin{bmatrix} 1 & 0 \\ -2 & -3 \end{bmatrix}$, $P = \begin{bmatrix} -2 \\ 5 \end{bmatrix}$

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Show your working

Teacher answers & working

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S1

Find the image of each point under the matrix transformation.
$A = \begin{bmatrix} 0 & -2 \\ 1 & -3 \end{bmatrix}$, $P = \begin{bmatrix} -2 \\ -2 \end{bmatrix}$

Answer:
$\begin{bmatrix} 4 \\ 4 \end{bmatrix}$

Working:

Write the point as a column vector.

Multiply the transformation matrix by the point vector.
$AP = \begin{bmatrix} 0 & -2 \\ 1 & -3 \end{bmatrix} \begin{bmatrix} -2 \\ -2 \end{bmatrix}$.

Top entry = 0(-2) - 2(-2) = 4.

Bottom entry = 1(-2) - 3(-2) = 4.

The image is $\begin{bmatrix} 4 \\ 4 \end{bmatrix}$.

S2

Find the combined matrix for each pair of transformations.
A = $\begin{bmatrix} 2 & -2 \\ 1 & 3 \end{bmatrix}$, B = $\begin{bmatrix} 2 & 1 \\ 2 & 3 \end{bmatrix}$, apply A then B

Answer:
$\begin{bmatrix} 5 & -1 \\ 7 & 5 \end{bmatrix}$

Working:

If A is applied first and B is applied second, the combined matrix is BA.
$BA = \begin{bmatrix} 2 & 1 \\ 2 & 3 \end{bmatrix} \begin{bmatrix} 2 & -2 \\ 1 & 3 \end{bmatrix}$.

Top-left = 2(2) + 1(1) = 5.

Top-right = 2(-2) + 1(3) = -1.

Bottom-left = 2(2) + 3(1) = 7.

Bottom-right = 2(-2) + 3(3) = 5.

Combined matrix = $\begin{bmatrix} 5 & -1 \\ 7 & 5 \end{bmatrix}$.

Teacher answers & working

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S3

Find the fixed line for each transformation.

$A = \begin{bmatrix} 1 & -2 \\ 0 & 4 \end{bmatrix}$

Answer:

$y = 0$

Working:

A fixed point satisfies AX = X.

Let $X = \begin{bmatrix} x \\ y \end{bmatrix}$.

$$\begin{bmatrix} 1 & -2 \\ 0 & 4 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} x - 2y \\ 4y \end{bmatrix}.$$

Set corresponding entries equal: x - 2y = x and 4y = y.

From x - 2y = x, subtract x from both sides: -2y = 0.

So $y = 0$.

The fixed points lie on the line y = 0.

S4

Find the image of each point under the matrix transformation.

$A = \begin{bmatrix} 0 & -2 \\ 0 & 1 \end{bmatrix}$, $P = \begin{bmatrix} -4 \\ -2 \end{bmatrix}$

Answer:

$\begin{bmatrix} 4 \\ -2 \end{bmatrix}$

Working:

Write the point as a column vector.

Multiply the transformation matrix by the point vector.

$$AP = \begin{bmatrix} 0 & -2 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} -4 \\ -2 \end{bmatrix}.$$

Top entry = 0(-4) - 2(-2) = 4.

Bottom entry = 0(-4) + 1(-2) = -2.

The image is $\begin{bmatrix} 4 \\ -2 \end{bmatrix}$.

Teacher answers & working

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EA1

Find the combined matrix for each pair of transformations.

A = $\begin{bmatrix} 2 & 4 \\ 3 & 4 \end{bmatrix}$, B = $\begin{bmatrix} 3 & 1 \\ 1 & 0 \end{bmatrix}$, apply A then B

Answer:

$\begin{bmatrix} 9 & 16 \\ 2 & 4 \end{bmatrix}$

Working:

If A is applied first and B is applied second, the combined matrix is BA.

$\mathrm{BA} = \begin{bmatrix} 3 & 1 \\ 1 & 0 \end{bmatrix}\begin{bmatrix} 2 & 4 \\ 3 & 4 \end{bmatrix}$.

Top-left = 3(2) + 1(3) = 9.

Top-right = 3(4) + 1(4) = 16.

Bottom-left = 1(2) + 0(3) = 2.

Bottom-right = 1(4) + 0(4) = 4.

Combined matrix = $\begin{bmatrix} 9 & 16 \\ 2 & 4 \end{bmatrix}$.

EA2

Find the fixed line for each transformation.

$\mathrm{A} = \begin{bmatrix} 1 & 4 \\ 0 & 4 \end{bmatrix}$

Answer:

$\mathrm{y} = 0$

Working:

A fixed point satisfies AX = X.

Let $\mathrm{X} = \begin{bmatrix} x \\ y \end{bmatrix}$.

$\begin{bmatrix} 1 & 4 \\ 0 & 4 \end{bmatrix}\begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} x + 4y \\ 4y \end{bmatrix}$.

Set corresponding entries equal: x + 4y = x and 4y = y.

From x + 4y = x, subtract x from both sides: 4y = 0.

So $\mathrm{y} = 0$.

The fixed points lie on the line y = 0.

Teacher answers & working

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EB1

Find the image of each point under the matrix transformation.

$A = \begin{bmatrix} 1 & 0 \\ -2 & -3 \end{bmatrix}$, $P = \begin{bmatrix} -2 \\ 5 \end{bmatrix}$

Answer:

$\begin{bmatrix} -2 \\ -11 \end{bmatrix}$

Working:

Write the point as a column vector.

Multiply the transformation matrix by the point vector.

$AP = \begin{bmatrix} 1 & 0 \\ -2 & -3 \end{bmatrix} \begin{bmatrix} -2 \\ 5 \end{bmatrix}$.

Top entry = 1(-2) + 0(5) = -2.

Bottom entry = -2(-2) - 3(5) = -11.

The image is $\begin{bmatrix} -2 \\ -11 \end{bmatrix}$.